

LUCIDMSK CME PROGRAM

MODULE 09 — POST-TEST

Elbow MRI: Ligaments, Nerves, OCD & Tendon Injuries

Elbow · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)[™]

Read each question carefully and select the single best answer. A score of $\geq$ **70%** (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 UCL Grade III on coronal MRI:

- A Thickened; continuous fibers; edema
- B Partial discontinuity; T-sign
- C Complete fiber discontinuity; fluid gap
- D Calcification within the ligament

2 OCD of the capitellum — MRI finding indicating UNSTABLE lesion:

- A Intact cartilage; no fluid beneath
- B Subchondral edema only
- C T2 high signal BENEATH the fragment (undercutting) + cartilage breach
- D Stable low T1 signal; no articular breach

3 Lateral epicondylitis most commonly affects which structure?

- A Radial collateral ligament
- B Lateral ulnar collateral ligament (LUCL)
- C Common extensor tendon origin (ECRB most vulnerable)
- D Annular ligament

4 Ulnar nerve CSA measures 12 mm² at the cubital tunnel. Interpretation:

- A Normal — up to 20 mm² accepted
- B Borderline — repeat in 6 months
- C Abnormal — enlargement suggests cubital tunnel syndrome (normal ≤ 9 –10 mm²)
- D CSA is not meaningful on MRI

5 Best sequence and plane for distal biceps partial tear:

- A Coronal PD fat-sat
- B Sagittal T1
- C Axial PD fat-sat at the radial tuberosity
- D Coronal STIR

6 PLRI (posterolateral rotatory instability) is caused by disruption of:

- A Medial UCL anterior bundle
- B LUCL (lateral ulnar collateral ligament)
- C Annular ligament only
- D Common extensor tendon origin

7 Radiocapitellar joint space is 3 mm on coronal MRI. Significance:

- A Normal — 2–4 mm is accepted
- B Borderline
- C Narrowed — lateral compartment OA (normal 2–4 mm; concern <2 mm)
- D Always abnormal

8 Olecranon stress fracture in a baseball pitcher. Key MRI finding:

- A Complete fracture line on sagittal T1
- B Medial epicondyle avulsion
- C Transverse T1 hypointense/T2 hyperintense stress fracture line at posterior olecranon
- D Triceps tendon complete tear

9 UCL tear most commonly occurs at which location on elbow MRI?

- A Midsubstance of anterior bundle

- B Proximal insertion at medial epicondyle
- C Distal insertion at sublime tubercle
- D Posterior bundle only

10 Capitellar OCD most commonly affects:

- A Adults 30–50; tennis players
- B Adolescent athletes (10–15 years); overhead throwing or gymnastics
- C Elderly; overuse repetitive work
- D Young children 5–8 years; elbow fractures

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	C	C	C	C	C	B	A	C	B	B

Q1 UCL Grade III on coronal MRI:

Answer: C

A Thickened; continuous fibers; edema

B Partial discontinuity; T-sign

✓ **Complete fiber discontinuity; fluid gap**

D Calcification within the ligament

EXPLANATION Grade III UCL = complete disruption with fluid gap at the proximal or distal attachment. T-sign describes undersurface partial fiber injury. Grade II = partial discontinuity without complete gap.

Q2 OCD of the capitellum — MRI finding indicating UNSTABLE lesion:

Answer: C

A Intact cartilage; no fluid beneath

B Subchondral edema only

✓ **T2 high signal BENEATH the fragment (undercutting) + cartilage breach**

D Stable low T1 signal; no articular breach

EXPLANATION Unstable OCD: T2 fluid signal BENEATH the fragment (undercutting) + breach of overlying articular cartilage. Stable = intact cartilage, no fluid undercutting. Unstable → arthroscopic fixation or removal.

Q3 Lateral epicondylitis most commonly affects which structure?

Answer: C

A Radial collateral ligament

B Lateral ulnar collateral ligament (LUCL)

✓ **Common extensor tendon origin (ECRB most vulnerable)**

D Annular ligament

EXPLANATION Lateral epicondylitis (tennis elbow) = tendinopathy/tear at the common extensor origin, with ECRB most vulnerable. The LUCL can co-tear but is a separate structure.

Q4 Ulnar nerve CSA measures 12 mm² at the cubital tunnel. Interpretation:

Answer: C

A Normal — up to 20 mm² accepted

B Borderline — repeat in 6 months

✓ **Abnormal — enlargement suggests cubital tunnel syndrome (normal ≤9–10 mm²)**

D CSA is not meaningful on MRI

EXPLANATION Normal ulnar nerve CSA $\leq 9\text{--}10\text{ mm}^2$. $>10\text{--}12\text{ mm}^2$ = compression-related enlargement (cubital tunnel syndrome). MRI also shows T2 signal change and denervation in flexor carpi ulnaris.

Q5 Best sequence and plane for distal biceps partial tear:

Answer: C

- A Coronal PD fat-sat
- B Sagittal T1

✓ **Axial PD fat-sat at the radial tuberosity**

- D Coronal STIR

EXPLANATION Axial PD fat-sat at the radial tuberosity is most sensitive for distal biceps tears — especially posterior bursal surface partial tears seen best in cross-section. Axial is primary; sagittal/coronal supplement.

Q6 PLRI (posterolateral rotatory instability) is caused by disruption of:

Answer: B

- A Medial UCL anterior bundle

✓ **LUCL (lateral ulnar collateral ligament)**

- C Annular ligament only
- D Common extensor tendon origin

EXPLANATION PLRI = LUCL disruption. The LUCL is the primary stabilizer against posterolateral rotatory subluxation. Pivot-shift test of the elbow is positive. Coronal MRI shows LUCL discontinuity/laxity.

Q7 Radiocapitellar joint space is 3 mm on coronal MRI. Significance:

Answer: A

✓ **Normal — 2–4 mm is accepted**

- B Borderline
- C Narrowed — lateral compartment OA (normal 2–4 mm; concern $<2\text{ mm}$)
- D Always abnormal

EXPLANATION Normal radiocapitellar joint space is 2–4 mm. 3 mm is within normal range. Values $<2\text{ mm}$ suggest narrowing. Values $>5\text{ mm}$ suggest lateral distension or soft tissue interposition.

Q8 Olecranon stress fracture in a baseball pitcher. Key MRI finding:

Answer: C

- A Complete fracture line on sagittal T1
- B Medial epicondyle avulsion

✓ **Transverse T1 hypointense/T2 hyperintense stress fracture line at posterior olecranon**

- D Triceps tendon complete tear

EXPLANATION Olecranon stress fractures in throwers: transverse stress fracture line at posterior olecranon (physis in adolescents; proximal diaphysis in adults) — T1 low / T2 STIR high with surrounding marrow edema.

Q9 UCL tear most commonly occurs at which location on elbow MRI?

Answer: B

A Midsubstance of anterior bundle

✓ **Proximal insertion at medial epicondyle**

C Distal insertion at sublime tubercle

D Posterior bundle only

EXPLANATION UCL tears most commonly occur at the PROXIMAL insertion at the medial epicondyle in throwing athletes. Distal (sublime tubercle) tears are second. Assess the anterior bundle specifically on coronal images.

Q10 Capitellar OCD most commonly affects:

Answer: B

A Adults 30–50; tennis players

✓ **Adolescent athletes (10–15 years); overhead throwing or gymnastics**

C Elderly; overuse repetitive work

D Young children 5–8 years; elbow fractures

EXPLANATION Capitellar OCD predominantly affects adolescent athletes (10–15 years) in overhead sports (baseball, gymnastics). Lateral compartment undergoes compressive shear. Early MRI diagnosis is critical for treatment planning.